

while the last null character is for specifying the end of the sequence. In the second case, by using string literals, the null character (`\0`) is automatically inserted at the end.

Applying Null-terminated Sequence Of Characters

Null terminated sequence of characters is an effective way of treating strings. They can also be used in procedures. For example, the extraction and insertion operators `cin` and `cout` support these sequences and can be used directly to extract strings of characters or to insert them. The following program illustrates this.

```
// null-terminated sequences of characters
#include <iostream.h>
int main ()
{
    char q1[] = "Please, enter your first name:
";
    char g1[] = "Hello, ";
    char y1[ 80];
    cout << q1;
    cin >> y1;
    cout << g1 << y1 << "!";
    return 0;
}
```

The output of this program is:

```
Please, enter your first name: Harry
Hello, Harry
```

In the above program, three arrays of type `char` are declared. They are `q1`, `g1` and `y1`, respectively. The first two arrays `q1` and `g1` are initialised with string literals constant, while the third one `y1` is left uninitialised. For the first two arrays `q1` and `g1`, the size was implicitly defined, while for the last one, the size is explicitly defined - 80 characters. When we run the program, it prompts you to enter your name. The name